

Five-String Acoustic Resonator — Rev A

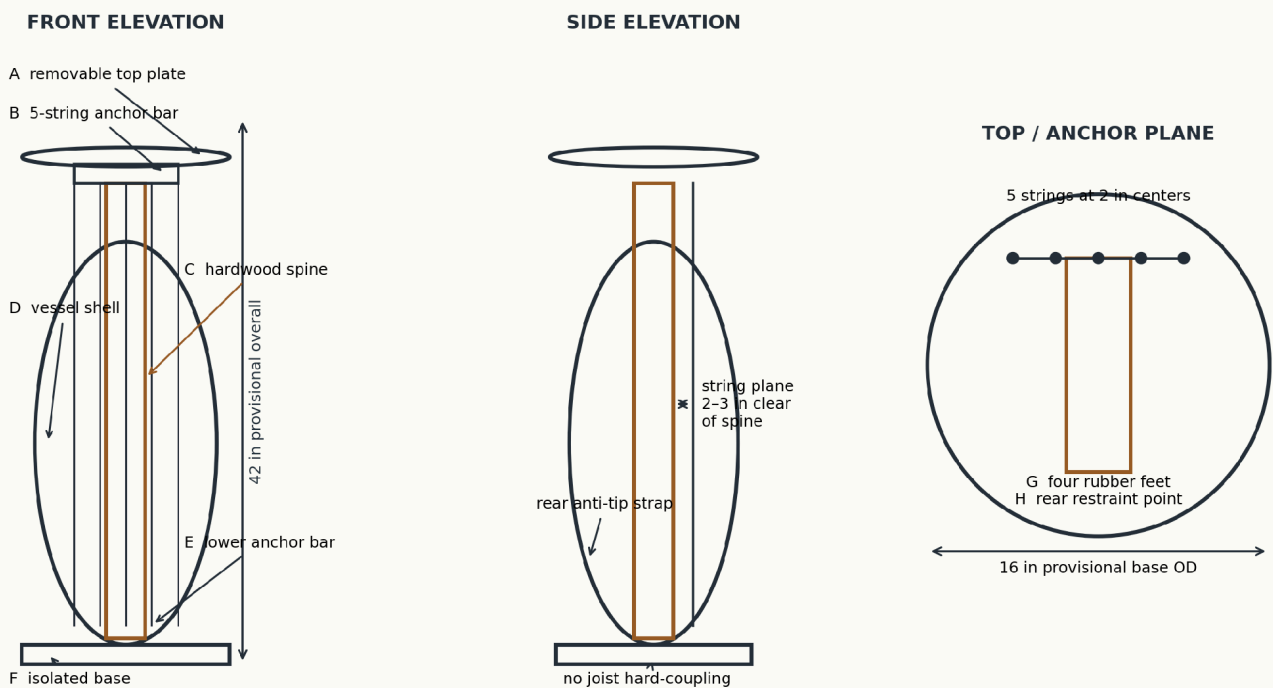
Build-ready first-pass schematic • acoustic/passive baseline • 13 August 2026

Design status: PROVISIONAL until the measurement gates in §4 are completed.

Build intent	Hard boundary	First-pass criterion
Upright five-string resonator using a vessel as an acoustic enclosure and a central hardwood load path.	No high voltage, energized helix, “Tesla coil,” exposed mains, smoke, salt, or mechanism inferred from mythology.	Stable, serviceable, audibly resonant, and measurable before any powered excitation is added.

1. Labeled geometry

REV A — ACOUSTIC-ONLY REFERENCE GEOMETRY (NOT TO SCALE)



Geometry rule. Strings occupy one serviceable plane in front of the spine. The vessel is an enclosure and guard, not a primary structural member. All string tension closes through the upper anchor, hardwood spine, lower anchor, and base frame.

2. Provisional dimensions and acceptance limits

Feature	Rev A nominal	Allowed / gate	Design note
Overall height	42 in (1067 mm)	36–48 in	Must fit room and remain tip-stable
Base outside diameter	16 in (406 mm)	≥ 14 in	Wider if total center of gravity is high
Vessel maximum OD / clear ID	14 / ≥ 12 in	Measure actual	≥ 1 in clearance to moving strings/hardware
Top plate diameter	16 in	14–18 in	Thin metal or 1/4–3/8 in plywood; removable
Spine	2 × 3 in hardwood, 40 in long	Engineer to load	Straight grain; no cracks/large knots

Feature	Rev A nominal	Allowed / gate	Design note
Speaking string length L	32 in (813 mm)	28–36 in	All five equal within 1/16 in unless intentionally scaled
String spacing	2 in centers	1.5–2.25 in	Enough access for plucking and damping
String-to-spine clearance	2–3 in	≥ 1.5 in	No contact at full excursion
Top acoustic gap	1.5 in	1–2 in	Open annulus; no pinching of airflow
Foot pattern	4 feet on 13–14 in circle	As base permits	All feet loaded; no rocking
Initial tension limit	≤ 20 lbf/string; ≤ 100 lbf total	Mandatory Rev A cap	Verify with string maker/tension calculator

3. Parts list

ID	Part	Qty	Specification	Safety / assembly note
P1	Base frame	1	3/4 in Baltic-birch plywood, laminated to 1.5 in total, 16 in OD	No cracked, warped, or particle-board base
P2	Central spine	1	Hard maple/ash, nominal 2 × 3 × 40 in	Primary load path
P3	Upper/lower anchor bars	2	1/4 in steel bar or 3/8 in aluminum, 8 × 2 in	Deburr; five holes at 2 in centers
P4	Tuning machines	5	Enclosed bass or zither pins rated for chosen strings	Mount to separate head block if needed
P5	Strings	5	Low-tension acoustic strings selected by calculated tension	Start ≤20 lbf each; install eye protection
P6	Bridge / nut strips	2	Hardwood, bone, or acetal; replaceable	Rounded bearing edges
P7	Vessel shell	1	Glass/acrylic/wood vessel, ≥12 in clear ID	Not load bearing; restrained with padded clamps
P8	Top plate	1	Thin cymbal-like metal plate or 1/4–3/8 in plywood, 14–18 in	Insulated from shell; removable
P9	Base feet	4	Neoprene/Sorbothane feet, 1–1.5 in OD, through-bolted	Non-slip; equal height
P10	Vessel clamps	3–4	Padded stainless straps or wooden cleats	No point loading on glass
P11	Anti-tip restraint	1	Furniture strap/cable rated ≥200 lbf	Attach to wall stud, not drywall alone
P12	Guards/covers	As needed	Polycarbonate shields over anchors/tuners	Contain a broken string end
P13	Fasteners	Set	Grade-marked machine bolts, washers, nyloc nuts	No wood screws in primary tension path
P14	Optional pickup	1	Passive piezo/contact microphone	Electrically isolated; removable
P15	Optional exciter — Phase 2 only	1	12–24 VDC, ≤30 W tactile exciter + fused isolated supply	Not installed until acoustic baseline passes

4. Dimensions still required before cutting or tensioning

- Room clear height and the exact footprint available, including walking clearance.
- Vessel: total height, maximum outside diameter, mouth inside/outside diameter, wall thickness, base shape, material, mass, and any chips or cracks.
- Chosen top plate/cymbal: diameter, mass, center-hole diameter, edge condition, and mounting hardware.
- Chosen string set: material, gauge/core type, unit weight or manufacturer tension at proposed pitch and 32 in scale, plus safe tuning range.
- Tuner footprint, post diameter, mounting thickness, and required access for winding.
- Spine stock actual dimensions, species, grain direction, straightness, and defects.
- Finished assembly mass and estimated center-of-gravity height; perform the 10° tip test before playing.
- Wall-stud location for anti-tip restraint; electrical outlet and cable route if Phase 2 is attempted.
- Target pitches and permitted sound-pressure level at the operator position.

4A. Drive measurement cross-check

Source record:  Pentahelix.

Room plan — 13 ft × 12.5 ft overall floor dimensions — **PARTIAL**. This does not establish clear height, exact installation footprint, walking clearance, stud location, outlet, or cable route.

Vessel — **UNKNOWN**: total height, maximum outside diameter, mouth inside/outside diameter, wall thickness, base shape, material, mass, and chip/crack condition.

Top plate/cymbal — **UNKNOWN**: diameter, mass, center-hole diameter, edge condition, and mounting hardware.

Strings — **UNKNOWN**: selected set, material, gauge/core type, unit weight or manufacturer tension at the exact scale and target pitch, and safe tuning range.

Tuners — **UNKNOWN**: model/rating, footprint, post diameter, mounting thickness, and winding access.

Spine stock — **UNKNOWN**: actual dimensions, species, grain direction, straightness, and defects.

Finished assembly/restraint — **UNKNOWN**: assembly mass, center-of-gravity height, wall-stud location, and 10° tip-test result.

Target pitches and permitted operator-position SPL — **UNKNOWN**.

Build disposition — **HOLD**. The 42 × 16 in envelope and 32 in speaking length remain provisional, not cleared for cutting or tensioning. Retaining them requires actual vessel clear ID ≥12 in, actual five-string tension ≤20 lbf per string at the exact scale and target pitches, verified component ratings, and the unresolved room/installation checks.

Tension gate: For each string, calculate or obtain manufacturer tension at the exact scale and target pitch. Sum all five. Do not exceed the lowest rating of the string, tuner, anchor, spine, fastener, or base connection; apply at least a 4× safety factor to anchors and restraint hardware. If ratings are unknown, do not tension.

5. Safe mounting and assembly

1. Dry-fit the base, spine, upper head block, and lower anchor without the vessel. Through-bolt the spine into a cross-braced base; use large washers or steel backing plates.
2. Fit four isolation feet and confirm all four carry load. Add the rear anti-tip strap to a structural wall stud, leaving only enough slack for isolation movement.
3. Mount the vessel independently with three or four padded restraints. Maintain a continuous 1 in minimum clearance from all string and anchor hardware. Never clamp glass at a single point or drill tempered glass.
4. Install rounded nut/bridge strips and guarded anchor bars. Orient cut string ends inward and cover them with polycarbonate or a fastened guard.
5. Install strings one at a time at hand-snug tension. Stand out of the string plane, wear eye protection, and increase in small equal increments while watching base lift, spine bow, fastener movement, and vessel clearance.
6. At ≤25%, 50%, 75%, and 100% of the Rev A tension cap, stop and inspect. Abort on cracking, creaking, permanent deflection, rocking, loosening, or contact with the vessel.
7. Fit the top plate on an insulated, captive mount only after the string frame is stable. The top must not be able to lift free during vibration.

6. Airflow and electrical constraints

Domain	Allowed in Rev A	Constraint / exclusion
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Airflow	Natural convection; one guarded external fan for mapping only	Fan stays outside vessel, ≥ 12 in from strings and loose fibers. No airflow through electronics; no fan-induced string contact.
Flow visualization	Short yarn tufts or lightweight tissue streamers	No smoke, incense, combustion, powders, salt mist, aerosols, or material that can enter lungs/electronics.
Copper helix	Decorative, mechanically secured, electrically floating	Not required for acoustic operation. It must not touch strings, tuners, pickup wiring, top plate, or electrical ground.
Acoustic sensing	Passive contact mic/piezo into battery recorder	Secure cable strain relief; avoid trip loops.
Powered excitation	Excluded from baseline; Phase 2 may use 12–24 VDC, ≤ 30 W	Isolated listed supply outside vessel, fused at source, guarded connector, no exposed conductors, no mains inside assembly, and never > 48 VDC.
Subwoofer/structure	External airborne excitation at modest level	No bolting or hard-coupling to house joists in Rev A. Keep loose objects clear and monitor floor/wall vibration.
Sound exposure	Start quiet; increase only as needed	Use hearing protection and measured SPL; stop for pain, pressure, dizziness, or rattling building components.

7. Commissioning and test sequence

Test	Method	Pass / abort criterion
T0 — Inspection	Photograph joints; torque-mark nuts; confirm vessel clearance and guards.	No cracks, rocking, loose hardware, sharp edges, or unguarded string ends.
T1 — Static proof	At the Rev A total tension cap, leave for 30 min; measure spine bow and base lift before/after.	No progressive movement; residual deflection ≤ 1 mm after unloading. Any audible crack/creak = fail.
T2 — Tip/restraint	With strings damped, tilt or shim base to 10° in the least-stable direction; verify restraint.	Assembly does not overturn; feet remain controlled; strap/anchor undamaged.
T3 — Tap map	Strings damped, tap base, vessel, spine, and top separately; record phone spectrum and decay.	Repeatable peaks; no buzz caused by loose hardware.
T4 — Individual strings	Pluck one string at a time, low amplitude; record fundamental, harmonics, and 10 dB decay time.	Clear tone, no vessel strike, anchor slip, or growing detune.
T5 — Coupling matrix	For each string, measure response at spine, vessel, top plate, and base with contact mic.	Record amplitude ratios; do not interpret analogy as cause.
T6 — Airflow map	Use yarn tufts around top gap with no powered exciter; video from fixed angle. Then repeat with one guarded external fan.	Tufts remain clear of strings/hardware; no debris ingestion or unstable vibration.
T7 — Room sweep	Place an external speaker/subwoofer ≥ 3 ft away; sweep 20–200 Hz slowly at low SPL, strings initially damped.	Stop on structural rattles, visible motion, pain/pressure, or SPL limit. Log resonances, not “field” claims.
T8 — Optional Phase 2	Only after T0–T7 pass: attach ≤ 30 W low-voltage exciter to base, fuse it, repeat T3–T7 at minimum drive.	No temperature rise $> 10^\circ\text{C}$ at exciter mount after 10 min; no loosening or new contact.

8. Evidence log template

For every run record: date/time • configuration revision • string gauges/pitches/tensions • sensor/phone position • excitation source and level • room doors/windows/fans • peak frequencies • decay times • visible motion • temperature • anomalies • stop reason. Change one variable per run.

9. Mechanical comparison with separate mythic analogy

Interpretation boundary. The analogy column is cultural/design language only. It supplies no evidence of acoustic, electrical, structural, physiological, or environmental mechanism.

Physical feature	Mythic analogy only	Engineering interpretation / boundary
Long central spine + resonating enclosure	Veena / vina	Useful visual family: long neck and resonant body; actual mechanism is ordinary string and enclosure vibration.
Five vertical strings	Lyre / harp	Useful layout family: individually tuned strings; actual behavior depends on length, tension, mass per length, anchors, and damping.
Ordered spacing and removable top	Apollo's lyre	Proportion/order analogy only; no solar, healing, or field claim.
Environmental listening / room sweep	Dagda's harp; Orpheus's lyre	Narrative analogy for music affecting a setting; measured room modes and structural coupling are the only testable subjects here.
Open top gap and airflow mapping	Gandharvas / Aeolian-harp family	Wind/breath analogy only; airflow is visualized with tufts and measured as ordinary fluid motion.
Optional unenergized copper ornament	Hermes/caduceus imagery	Decorative staff/serpent analogy only. No coil, antenna, charge-focusing, or electromagnetic-mechanism claim.

10. Rev A exclusions and change-control gate

- No Tesla coil, primary/secondary HV winding, capacitive terminal, corona, exposed energized metal, or predicted electromagnetic “field pattern.”
- No claim that a cymbal “focuses charge,” that a helix improves field stability, or that mythology identifies a mechanism.
- No structural attachment to joists, walls, or building framing except a slack anti-tip safety restraint to a wall stud.
- No smoke, flame, incense, salt, conductive liquids, dust, or aerosol near the assembly.
- Only one change per revision. A powered exciter, different string family, hard structural coupling, or energized conductive element requires a new risk review—not a silent Rev A modification.

11. First build decision

Recommended cut plan after measurements: retain the 42 × 16 in envelope and 32 in speaking length if the actual vessel supplies ≥12 in clear ID and the selected five strings remain at or below 20 lbf each. Otherwise resize the frame around the vessel and recalculate every string tension before cutting anchor bars.